

Warm-Up: Thinking in Classes

Today we'll practice breaking down real-world problems into pieces of software. Don't worry about being "right." The goal is to *practice thinking like a programmer*.

How to do this activity

1. Read each short scenario and its requirements. Your group picks ONE.
 2. Brainstorm: *What classes (nouns) might we need?*
 - Example: A library system might need a Book class and a Patron class.
 3. EXTRA: Do you think any of the classes might be related to one another? How?
 4. EXTRA: Pick **one** class you listed and design it.
 - Give it some properties (with Java data types).
 - List a few methods it might have (just names are fine for now).
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Scenarios

1) Public Library Checkout

- A patron can check out up to 10 books.
 - Books have authors, copies, and due dates.
 - Late returns cost money.
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2) Fitness Class Sign-Ups

- Members can sign up for classes (like yoga, spin, pilates).
 - Each class has a max number of spots.
 - Instructors teach classes at set times.
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3) Food Delivery App

- Customers order food from restaurants.
 - Drivers deliver orders.
 - Orders include multiple food items.
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4) School Grading System

- Students enroll in courses.
 - Teachers assign grades.
 - Courses have assignments and exams.
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5) Online Store

- Shoppers add items to a cart.
 - Each item has a price and description.
 - Orders can be paid with a credit card.
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6) Movie Theater Tickets

- People can buy tickets for a movie showing.
 - Each showing has a date, time, and theater room.
 - Tickets have seat numbers.
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7) Pet Adoption Center

- Animals are available for adoption.
 - People can apply to adopt.
 - Each animal has a name, species, and age.
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8) Ride-Sharing App

- Riders request rides.
 - Drivers accept rides.
 - Each ride has a start and end location.
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9) Hotel Booking

- Guests can reserve rooms.
 - Rooms have different types (single, double, suite).
 - Bookings have check-in and check-out dates.
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10) Music Playlist App

- Users create playlists.
 - A playlist can contain many songs.
 - Songs have titles, artists, and lengths.
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Extra Credit Example

Pick one class you listed and “design” it:

Example: Book

- Properties:
 - String title
 - String author
 - boolean isCheckedOut
 - LocalDate dueDate
- Methods:
 - checkOut()
 - returnBook()

- `renew()`

👉 Your turn: Pick a class from one of the scenarios and sketch out its properties and methods!